DBMS MINI PROJECT

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Section: 5K

Project title: Digital Museum Artifact & Restoration Management System

DDL Statements:

```
mysql> CREATE DATABASE digital_museum_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE digital_museum_db;
Database changed
```

```
mysql> CREATE TABLE MUSEUM (
    ->     museum_id INT AUTO_INCREMENT,
    ->     museum_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     address VARCHAR(255),
    ->     city VARCHAR(100),
    ->     state VARCHAR(100),
    ->     country VARCHAR(100),
    ->     contact_email VARCHAR(150),
    ->     contact_phone VARCHAR(30),
    ->     established_year YEAR,
    ->
    ->     PRIMARY KEY (museum_id)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE COLLECTION (
    ->     collection_id INT AUTO_INCREMENT,
    ->     museum_id INT NOT NULL,
    ->     collection_name VARCHAR(150) NOT NULL,
    ->     description TEXT,
    ->     historical_period VARCHAR(100),
    ->     collection_type VARCHAR(100),
    ->     date_created DATE,
    ->
    ->     PRIMARY KEY (collection_id),
    ->
    ->     CONSTRAINT fk_collection_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE CREATOR (
    ->     creator_id INT NOT NULL,
    ->     creator_name VARCHAR(150) NOT NULL,
    ->     birth_year YEAR,
    ->     nationality VARCHAR(100),
    ->     biography TEXT,
    ->
    ->     PRIMARY KEY (creator_id)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE ARTIFACT (
    ->     artifact_id INT AUTO_INCREMENT,
    ->     collection_id INT NOT NULL,
    ->     creator_id INT,
    ->     accession_number VARCHAR(100) NOT NULL,
    ->     artifact_name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     artifact_type VARCHAR(100),
    ->     origin_place VARCHAR(150),
    ->     creation_date DATE,
    ->     estimated_value DECIMAL(15,2),
    ->     acquisition_date DATE,
    ->     status VARCHAR(100),
    ->
    ->     PRIMARY KEY (artifact_id),
    ->
    ->     CONSTRAINT uq_artifact_accession UNIQUE (accession_number),
    ->
    ->     CONSTRAINT fk_artifact_collection FOREIGN KEY (collection_id) REFERENCES COLLECTION(collection_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    ->
    ->     CONSTRAINT fk_artifact_creator FOREIGN KEY (creator_id) REFERENCES CREATOR(creator_id) ON UPDATE CASCADE ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE STAFF (
    ->     staff_id INT AUTO_INCREMENT,
    ->     staff_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150) NOT NULL,
    ->     phone VARCHAR(30),
    ->     designation VARCHAR(100),
    ->     department VARCHAR(100),
    ->     username VARCHAR(100) NOT NULL,
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     staff_status VARCHAR(50),
    ->
    ->     PRIMARY KEY (staff_id),
    ->
    ->     CONSTRAINT uq_staff_email
    ->         UNIQUE (email),
    ->
    ->     CONSTRAINT uq_staff_username
    ->         UNIQUE (username)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE EXHIBITION (
    ->     exhibition_id INT NOT NULL,
    ->     museum_id INT NOT NULL,
    ->     exhibition_name VARCHAR(200) NOT NULL,
    ->     description TEXT,
    ->     theme VARCHAR(150),
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->
    ->     PRIMARY KEY (exhibition_id),
    ->
    ->     CONSTRAINT fk_exhibition_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE DISPLAY (
    ->     exhibition_id INT NOT NULL,
    ->     artifact_id INT NOT NULL,
    ->     display_start_date DATE,
    ->     display_end_date DATE,
    ->     display_location VARCHAR(150),
    ->     display_case VARCHAR(100),
    ->     display_notes TEXT,
    ->
    ->     CONSTRAINT fk_display_exhibition
    ->         FOREIGN KEY (exhibition_id)
    ->         REFERENCES EXHIBITION(exhibition_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_display_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE ARTIFACT_MEDIA (
    ->     media_id INT NOT NULL,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     media_type VARCHAR(50),
    ->     file_name VARCHAR(255),
    ->     file_url VARCHAR(500),
    ->     description TEXT,
    ->     uploaded_on DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     file_size BIGINT,
    ->
    ->     PRIMARY KEY (media_id),
    ->
    ->     CONSTRAINT fk_media_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_media_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE ARTIFACT_HISTORY (
    ->     history_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     event_date DATE NOT NULL,
    ->     event_type VARCHAR(100),
    ->     location VARCHAR(150),
    ->     description TEXT,
    ->     source_reference VARCHAR(500),
    ->
    ->     PRIMARY KEY (history_id),
    ->
    ->     CONSTRAINT fk_history_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_history_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE RESTORATION_PROJECT (
    ->     project_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     manager_id INT,
    ->     project_name VARCHAR(200) NOT NULL,
    ->     objective TEXT,
    ->     start_date DATE,
    ->     end_date DATE,
    ->     status VARCHAR(50),
    ->     cost DECIMAL(15,2),
    ->
    ->     PRIMARY KEY (project_id),
    ->
    ->     CONSTRAINT fk_project_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_manager
    ->         FOREIGN KEY (manager_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE CONDITION_ASSESSMENT (
    ->     assessment_id INT AUTO_INCREMENT,
    ->     artifact_id INT NOT NULL,
    ->     staff_id INT,
    ->     assessment_date DATE NOT NULL,
    ->     conditions VARCHAR(100),
    ->     damage_type VARCHAR(100),
    ->     damage_description TEXT,
    ->     recommended_action TEXT,
    ->     priority VARCHAR(50),
    ->
    ->     PRIMARY KEY (assessment_id),
    ->
    ->     CONSTRAINT fk_assessment_artifact
    ->         FOREIGN KEY (artifact_id)
    ->         REFERENCES ARTIFACT(artifact_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_assessment_staff
    ->         FOREIGN KEY (staff_id)
    ->         REFERENCES STAFF(staff_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE SET NULL
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE RESTORATION_EXPERT (
    ->     expert_id INT AUTO_INCREMENT,
    ->     expert_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     hours_worked DECIMAL(8,2),
    ->
    ->     PRIMARY KEY (expert_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE PROJECT_EXPERT (
    ->     project_id INT NOT NULL,
    ->     expert_id INT NOT NULL,
    ->
    ->     CONSTRAINT fk_project_expert_project
    ->         FOREIGN KEY (project_id)
    ->         REFERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_expert_expert
    ->         FOREIGN KEY (expert_id)
    ->         REFERENCES RESTORATION_EXPERT(expert_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE RESTORATION_MATERIAL (
    ->     material_id INT AUTO_INCREMENT,
    ->     material_name VARCHAR(150) NOT NULL,
    ->     material_type VARCHAR(100),
    ->     supplier VARCHAR(150),
    ->     unit VARCHAR(50),
    ->     unit_cost DECIMAL(12,2),
    ->     quantity_used DECIMAL(10,2),
    ->     total_cost DECIMAL(15,2),
    ->
    ->     PRIMARY KEY (material_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE PROJECT_MATERIAL (
    ->     project_id INT NOT NULL,
    ->     material_id INT NOT NULL,
    ->
    ->     CONSTRAINT fk_project_material_project
    ->         FOREIGN KEY (project_id)
    ->         REFERENCES RESTORATION_PROJECT(project_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_project_material_material
    ->         FOREIGN KEY (material_id)
    ->         REFERENCES RESTORATION_MATERIAL(material_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE VISITOR (
    ->     visitor_id INT AUTO_INCREMENT,
    ->     visitor_name VARCHAR(150) NOT NULL,
    ->     email VARCHAR(150),
    ->     phone VARCHAR(30),
    ->     age INT,
    ->     nationality VARCHAR(100),
    ->     registration_date DATE,
    ->
    ->     PRIMARY KEY (visitor_id)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE VISIT_RECORD (
    ->     visit_id INT AUTO_INCREMENT,
    ->     visitor_id INT NOT NULL,
    ->     museum_id INT NOT NULL,
    ->     visit_date DATE NOT NULL,
    ->     entry_time TIME,
    ->     exit_time TIME,
    ->     visit_type VARCHAR(50),
    ->     ticket_type VARCHAR(50),
    ->
    ->     PRIMARY KEY (visit_id),
    ->
    ->     CONSTRAINT fk_visit_visitor
    ->         FOREIGN KEY (visitor_id)
    ->         REFERENCES VISITOR(visitor_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE,
    ->
    ->     CONSTRAINT fk_visit_museum
    ->         FOREIGN KEY (museum_id)
    ->         REFERENCES MUSEUM(museum_id)
    ->         ON UPDATE CASCADE
    ->         ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> show tables;
+-----------------------------+
| Tables_in_digital_museum_db |
+-----------------------------+
| artifact                    |
| artifact_history            |
| artifact_media              |
| collection                  |
| condition_assessment        |
| creator                     |
| display                     |
| exhibition                  |
| museum                      |
| project_expert              |
| project_material            |
| restoration_expert          |
| restoration_material        |
| restoration_project         |
| staff                       |
| visit_record                |
| visitor                     |
+-----------------------------+
17 rows in set (0.02 sec)
```

```
mysql> desc artifact;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| artifact_id      | int           | NO   | PRI | NULL    | auto_increment |
| collection_id    | int           | NO   | MUL | NULL    |                |
| creator_id       | int           | YES  | MUL | NULL    |                |
| accession_number | varchar(100)  | NO   | UNI | NULL    |                |
| artifact_name    | varchar(200)  | NO   |     | NULL    |                |
| description      | text          | YES  |     | NULL    |                |
| artifact_type    | varchar(100)  | YES  |     | NULL    |                |
| origin_place     | varchar(150)  | YES  |     | NULL    |                |
| creation_date    | date          | YES  |     | NULL    |                |
| estimated_value  | decimal(15,2) | YES  |     | NULL    |                |
| acquisition_date | date          | YES  |     | NULL    |                |
| status           | varchar(100)  | YES  |     | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
12 rows in set (0.01 sec)

mysql> desc artifact_history;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| history_id       | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id      | int          | NO   | MUL | NULL    |                |
| staff_id         | int          | YES  | MUL | NULL    |                |
| event_date       | date         | NO   |     | NULL    |                |
| event_type       | varchar(100) | YES  |     | NULL    |                |
| location         | varchar(150) | YES  |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| source_reference | varchar(500) | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
mysql> desc artifact_media;
+-------------+-------------+------+-----+-------------------+-------------------+
| Field       | Type        | Null | Key | Default           | Extra             |
+-------------+-------------+------+-----+-------------------+-------------------+
| media_id    | int         | NO   | PRI | NULL              |                   |
| artifact_id | int         | NO   | MUL | NULL              |                   |
| staff_id    | int         | YES  | MUL | NULL              |                   |
| media_type  | varchar(50) | YES  |     | NULL              |                   |
| file_name   | varchar(255)| YES  |     | NULL              |                   |
| file_url    | varchar(500)| YES  |     | NULL              |                   |
| description | text        | YES  |     | NULL              |                   |
| uploaded_on | datetime    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| file_size   | bigint      | YES  |     | NULL              |                   |
+-------------+-------------+------+-----+-------------------+-------------------+
9 rows in set (0.01 sec)

mysql> desc collection;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| collection_id     | int          | NO   | PRI | NULL    | auto_increment |
| museum_id         | int          | NO   | MUL | NULL    |                |
| collection_name   | varchar(150) | NO   |     | NULL    |                |
| description       | text         | YES  |     | NULL    |                |
| historical_period | varchar(100) | YES  |     | NULL    |                |
| collection_type   | varchar(100) | YES  |     | NULL    |                |
| date_created      | date         | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
```

```
mysql> desc condition_assessment;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| assessment_id      | int          | NO   | PRI | NULL    | auto_increment |
| artifact_id        | int          | NO   | MUL | NULL    |                |
| staff_id           | int          | YES  | MUL | NULL    |                |
| assessment_date    | date         | NO   |     | NULL    |                |
| conditions         | varchar(100) | YES  |     | NULL    |                |
| damage_type        | varchar(100) | YES  |     | NULL    |                |
| damage_description | text         | YES  |     | NULL    |                |
| recommended_action | text         | YES  |     | NULL    |                |
| priority           | varchar(50)  | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)

mysql> desc crator;
ERROR 1146 (42S02): Table 'digital_museum_db.crator' doesn't exist
mysql> desc creator;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| creator_id   | int          | NO   | PRI | NULL    |       |
| creator_name | varchar(150) | NO   |     | NULL    |       |
| birth_year   | year         | YES  |     | NULL    |       |
| nationality  | varchar(100) | YES  |     | NULL    |       |
| biography    | text         | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> desc display;
+--------------------+--------------+------+-----+---------+-------+
| Field              | Type         | Null | Key | Default | Extra |
+--------------------+--------------+------+-----+---------+-------+
| exhibition_id      | int          | NO   | MUL | NULL    |       |
| artifact_id        | int          | NO   | MUL | NULL    |       |
| display_start_date | date         | YES  |     | NULL    |       |
| display_end_date   | date         | YES  |     | NULL    |       |
| display_location   | varchar(150) | YES  |     | NULL    |       |
| display_case       | varchar(100) | YES  |     | NULL    |       |
| display_notes      | text         | YES  |     | NULL    |       |
+--------------------+--------------+------+-----+---------+-------+
7 rows in set (0.01 sec)

mysql> desc exhibition;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| exhibition_id   | int          | NO   | PRI | NULL    |       |
| museum_id       | int          | NO   | MUL | NULL    |       |
| exhibition_name | varchar(200) | NO   |     | NULL    |       |
| description     | text         | YES  |     | NULL    |       |
| theme           | varchar(150) | YES  |     | NULL    |       |
| start_date      | date         | YES  |     | NULL    |       |
| end_date        | date         | YES  |     | NULL    |       |
| status          | varchar(50)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

```
mysql> desc museum;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| museum_id        | int          | NO   | PRI | NULL    | auto_increment |
| museum_name      | varchar(150) | NO   |     | NULL    |                |
| description      | text         | YES  |     | NULL    |                |
| address          | varchar(255) | YES  |     | NULL    |                |
| city             | varchar(100) | YES  |     | NULL    |                |
| state            | varchar(100) | YES  |     | NULL    |                |
| country          | varchar(100) | YES  |     | NULL    |                |
| contact_email    | varchar(150) | YES  |     | NULL    |                |
| contact_phone    | varchar(30)  | YES  |     | NULL    |                |
| established_year | year         | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
10 rows in set (0.01 sec)

mysql> desc project_expert;
+------------+------+------+-----+---------+-------+
| Field      | Type | Null | Key | Default | Extra |
+------------+------+------+-----+---------+-------+
| project_id | int  | NO   | MUL | NULL    |       |
| expert_id  | int  | NO   | MUL | NULL    |       |
+------------+------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> desc project_material;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| project_id  | int  | NO   | MUL | NULL    |       |
| material_id | int  | NO   | MUL | NULL    |       |
+-------------+------+------+-----+---------+-------+
2 rows in set (0.01 sec)
```

```
mysql> desc restoration_expert;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| expert_id    | int          | NO   | PRI | NULL    | auto_increment |
| expert_name  | varchar(150) | NO   |     | NULL    |                |
| email        | varchar(150) | YES  |     | NULL    |                |
| phone        | varchar(30)  | YES  |     | NULL    |                |
| hours_worked | decimal(8,2) | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.01 sec)

mysql> desc restoration_material;
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| material_id   | int           | NO   | PRI | NULL    | auto_increment |
| material_name | varchar(150)  | NO   |     | NULL    |                |
| material_type | varchar(100)  | YES  |     | NULL    |                |
| supplier      | varchar(150)  | YES  |     | NULL    |                |
| unit          | varchar(50)   | YES  |     | NULL    |                |
| unit_cost     | decimal(12,2) | YES  |     | NULL    |                |
| quantity_used | decimal(10,2) | YES  |     | NULL    |                |
| total_cost    | decimal(15,2) | YES  |     | NULL    |                |
+---------------+---------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)
```

```
mysql> desc restoration_project;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| project_id   | int           | NO   | PRI | NULL    | auto_increment |
| artifact_id  | int           | NO   | MUL | NULL    |                |
| manager_id   | int           | YES  | MUL | NULL    |                |
| project_name | varchar(200)  | NO   |     | NULL    |                |
| objective    | text          | YES  |     | NULL    |                |
| start_date   | date          | YES  |     | NULL    |                |
| end_date     | date          | YES  |     | NULL    |                |
| status       | varchar(50)   | YES  |     | NULL    |                |
| cost         | decimal(15,2) | YES  |     | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)

mysql> desc staff;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| staff_id      | int          | NO   | PRI | NULL    | auto_increment |
| staff_name    | varchar(150) | NO   |     | NULL    |                |
| email         | varchar(150) | NO   | UNI | NULL    |                |
| phone         | varchar(30)  | YES  |     | NULL    |                |
| designation   | varchar(100) | YES  |     | NULL    |                |
| department    | varchar(100) | YES  |     | NULL    |                |
| username      | varchar(100) | NO   | UNI | NULL    |                |
| password_hash | varchar(255) | NO   |     | NULL    |                |
| staff_status  | varchar(50)  | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

```
mysql> desc visitor;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| visitor_id        | int          | NO   | PRI | NULL    | auto_increment |
| visitor_name      | varchar(150) | NO   |     | NULL    |                |
| email             | varchar(150) | YES  |     | NULL    |                |
| phone             | varchar(30)  | YES  |     | NULL    |                |
| age               | int          | YES  |     | NULL    |                |
| nationality       | varchar(100) | YES  |     | NULL    |                |
| registration_date | date         | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)

mysql> desc visit_record;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| visit_id    | int         | NO   | PRI | NULL    | auto_increment |
| visitor_id  | int         | NO   | MUL | NULL    |                |
| museum_id   | int         | NO   | MUL | NULL    |                |
| visit_date  | date        | NO   |     | NULL    |                |
| entry_time  | time        | YES  |     | NULL    |                |
| exit_time   | time        | YES  |     | NULL    |                |
| visit_type  | varchar(50) | YES  |     | NULL    |                |
| ticket_type | varchar(50) | YES  |     | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

DML Statements:

```
mysql> INSERT INTO MUSEUM
    -> (museum_id, museum_name, description, address, city, state, country, contact_email, contact_phone, established_year)
    -> VALUES
    -> (1, 'National Heritage Museum',
    ->  'Museum preserving Indian historical and cultural artifacts.',
    ->  'MG Road', 'Bengaluru', 'Karnataka', 'India',
    ->  'info@nationalheritage.com', '08040001001', 1985),
    ->
    -> (2, 'Royal History Museum',
    ->  'Museum dedicated to royal history and traditional artifacts.',
    ->  'Palace Road', 'Mysuru', 'Karnataka', 'India',
    ->  'info@royalhistory.com', '08214000102', 1992),
    ->
    -> (3, 'Indian Art Museum',
    ->  'Museum showcasing traditional and modern Indian art.',
    ->  'Museum Road', 'New Delhi', 'Delhi', 'India',
    ->  'info@indianartmuseum.com', '01140001003', 1978),
    ->
    -> (4, 'South Indian Heritage Museum',
    ->  'Museum preserving cultural heritage from South India.',
    ->  'Temple Street', 'Chennai', 'Tamil Nadu', 'India',
    ->  'info@sihmuseum.com', '04440001004', 1995),
    ->
    -> (5, 'Ancient Civilisation Museum',
    ->  'Museum containing archaeological and ancient civilisation artifacts.',
    ->  'Heritage Avenue', 'Hyderabad', 'Telangana', 'India',
    ->  'info@ancientcivilisation.com', '04040001005', 2001);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO COLLECTION
    -> (collection_id, museum_id, collection_name, description, historical_period, collection_type, date_created)
    -> VALUES
    -> (1, 1, 'Ancient Indian Art',
    ->  'Collection of ancient Indian sculptures and artwork.',
    ->  'Ancient India', 'Sculpture', '1986-01-10'),
    ->
    -> (2, 1, 'Medieval Manuscripts',
    ->  'Collection of historical manuscripts and documents.',
    ->  'Medieval Period', 'Manuscript', '1987-05-15'),
    ->
    -> (3, 2, 'Royal Heritage',
    ->  'Artifacts belonging to historical royal families.',
    ->  'Medieval India', 'Royal Artifacts', '1993-03-20'),
    ->
    -> (4, 2, 'Mysuru Palace Collection',
    ->  'Objects associated with Mysuru royal history.',
    ->  '18th-20th Century', 'Royal Collection', '1994-04-15'),
    ->
    -> (5, 3, 'Indian Paintings',
    ->  'Collection of traditional Indian paintings.',
    ->  '19th-20th Century', 'Painting', '1979-02-12'),
    ->
    -> (6, 3, 'Modern Art',
    ->  'Collection of modern Indian artworks.',
    ->  '20th-21st Century', 'Modern Art', '1985-07-21'),
    ->
    -> (7, 4, 'Temple Artifacts',
    ->  'Artifacts collected from South Indian temples.',
    ->  'Ancient-Medieval', 'Religious Art', '1996-03-10'),
    ->
    -> (8, 4, 'Traditional Crafts',
    ->  'Traditional crafts and handmade cultural objects.',
    ->  '19th-20th Century', 'Craft', '1997-08-18'),
    ->
    -> (9, 5, 'Archaeological Collection',
    ->  'Artifacts discovered through archaeological research.',
    ->  'Ancient Period', 'Archaeological', '2002-01-20'),
    ->
    -> (10, 5, 'Coins and Currency',
    ->  'Historical coins and currency from different periods.',
    ->  'Ancient-Modern', 'Numismatics', '2003-06-11');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> ALTER TABLE CREATOR
    -> MODIFY birth_year INT;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO CREATOR
    -> (creator_id, creator_name, birth_year, nationality, biography)
    -> VALUES
    -> (1, 'Ravi Varma', 1848, 'Indian',
    ->  'Indian painter known for paintings based on Indian mythology.'),
    ->
    -> (2, 'Unknown Ancient Sculptor', 1200, 'Indian',
    ->  'Ancient sculptor whose identity remains unknown.'),
    ->
    -> (3, 'Mysuru Royal Artisan', 1750, 'Indian',
    ->  'Traditional artisan associated with the Mysuru royal court.'),
    ->
    -> (4, 'Unknown Manuscript Artist', 1600, 'Indian',
    ->  'Artist responsible for illustrating historical manuscripts.'),
    ->
    -> (5, 'Abanindranath Tagore', 1871, 'Indian',
    ->  'Indian painter and one of the founders of the Bengal School of Art.'),
    ->
    -> (6, 'Nandalal Bose', 1882, 'Indian',
    ->  'Indian painter associated with the Bengal School of Art.'),
    ->
    -> (7, 'Jamini Roy', 1887, 'Indian',
    ->  'Indian painter known for folk-inspired artistic styles.'),
    ->
    -> (8, 'Unknown Temple Artisan', 1400, 'Indian',
    ->  'Traditional artisan responsible for temple artwork.'),
    ->
    -> (9, 'Ancient Metalworker', 1100, 'Indian',
    ->  'Ancient artisan specializing in metal objects.'),
    ->
    -> (10, 'Royal Coin Maker', 1650, 'Indian',
    ->  'Artisan associated with historical royal coin production.');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO STAFF
    -> (staff_id, staff_name, email, phone, designation, department, username, password_hash, staff_status)
    -> VALUES
    -> (1, 'Anita Rao', 'anita@museum.com', '9876500001',
    ->  'Curator', 'Collections', 'anita.rao', 'hash_anita123', 'Active'),
    ->
    -> (2, 'Rahul Sharma', 'rahul@museum.com', '9876500002',
    ->  'Restoration Manager', 'Restoration', 'rahul.sharma', 'hash_rahul123', 'Active'),
    ->
    -> (3, 'Meera Nair', 'meera@museum.com', '9876500003',
    ->  'Archivist', 'Archives', 'meera.nair', 'hash_meera123', 'Active'),
    ->
    -> (4, 'Kiran Kumar', 'kiran@museum.com', '9876500004',
    ->  'Museum Director', 'Administration', 'kiran.kumar', 'hash_kiran123', 'Active'),
    ->
    -> (5, 'Priya Menon', 'priya@museum.com', '9876500005',
    ->  'Conservator', 'Restoration', 'priya.menon', 'hash_priya123', 'Active'),
    ->
    -> (6, 'Vivek Rao', 'vivek@museum.com', '9876500006',
    ->  'Researcher', 'Research', 'vivek.rao', 'hash_vivek123', 'Active'),
    ->
    -> (7, 'Sneha Iyer', 'sneha@museum.com', '9876500007',
    ->  'Curator', 'Exhibitions', 'sneha.iyer', 'hash_sneha123', 'Active'),
    ->
    -> (8, 'Arjun Das', 'arjun@museum.com', '9876500008',
    ->  'Security Officer', 'Security', 'arjun.das', 'hash_arjun123', 'Active'),
    ->
    -> (9, 'Divya Shah', 'divya@museum.com', '9876500009',
    ->  'Archivist', 'Archives', 'divya.shah', 'hash_divya123', 'Active'),
    ->
    -> (10, 'Manoj Kumar', 'manoj@museum.com', '9876500010',
    ->  'Technician', 'Digital Archives', 'manoj.kumar', 'hash_manoj123', 'Active');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO ARTIFACT
    -> (artifact_id, collection_id, creator_id, accession_number, artifact_name,
    ->  description, artifact_type, origin_place, creation_date,
    ->  estimated_value, acquisition_date, status)
    -> VALUES
    -> (1, 1, 2, 'ART-001',
    ->  'Ancient Stone Sculpture',
    ->  'Stone sculpture depicting an ancient Indian deity.',
    ->  'Sculpture', 'Hampi', '1250-01-01',
    ->  250000.00, '1986-03-12', 'Good'),
    ->
    -> (2, 1, 2, 'ART-002',
    ->  'Granite Temple Statue',
    ->  'Carved granite statue from a historic temple.',
    ->  'Sculpture', 'Hampi', '1300-01-01',
    ->  320000.00, '1987-04-15', 'Good'),
    ->
    -> (3, 2, 4, 'ART-003',
    ->  'Royal Manuscript',
    ->  'Handwritten manuscript containing historical records.',
    ->  'Manuscript', 'Mysuru', '1650-01-01',
    ->  150000.00, '1988-04-10', 'Good'),
    ->
    -> (4, 2, 4, 'ART-004',
    ->  'Historical Document',
    ->  'Historical administrative document.',
    ->  'Document', 'Karnataka', '1700-01-01',
    ->  80000.00, '1990-08-19', 'Good'),
    ->
    -> (5, 3, 3, 'ART-005',
    ->  'Royal Wooden Chest',
    ->  'Decorated wooden chest used by a historical royal family.',
    ->  'Furniture', 'Mysuru', '1780-01-01',
    ->  180000.00, '1994-02-15', 'Damaged'),
    ->
    -> (6, 3, 3, 'ART-006',
    ->  'Royal Sword',
    ->  'Ceremonial sword belonging to a royal collection.',
    ->  'Weapon', 'Mysuru', '1775-01-01',
    ->  300000.00, '1995-07-11', 'Good'),
    ->
    -> (7, 4, 3, 'ART-007',
    ->  'Royal Throne',
    ->  'Decorated wooden throne used for ceremonial purposes.',
    ->  'Furniture', 'Mysuru', '1800-01-01',
    ->  550000.00, '1996-01-20', 'Needs Restoration'),
    ->
    -> (8, 4, 3, 'ART-008',
    ->  'Royal Jewellery Box',
    ->  'Ornamented jewellery box from the royal collection.',
    ->  'Furniture', 'Mysuru', '1820-01-01',
    ->  210000.00, '1997-05-12', 'Good'),
    ->
    -> (9, 5, 1, 'ART-009',
    ->  'Mythological Painting',
    ->  'Painting depicting an Indian mythological scene.',
    ->  'Painting', 'Mysuru', '1890-01-01',
    ->  450000.00, '1987-06-20', 'Needs Restoration'),
    ->
    -> (10, 5, 5, 'ART-010',
    ->  'Bharat Mata Painting',
    ->  'Traditional Indian painting representing Bharat Mata.',
    ->  'Painting', 'Kolkata', '1905-01-01',
    ->  500000.00, '1990-02-11', 'Good'),
```

```
    ->
    -> (11, 5, 6, 'ART-011',
    ->  'Village Life Painting',
    ->  'Painting depicting rural Indian life.',
    ->  'Painting', 'Kolkata', '1920-01-01',
    ->  350000.00, '1995-03-22', 'Good'),
    ->
    -> (12, 6, 7, 'ART-012',
    ->  'Folk Art Composition',
    ->  'Folk-inspired modern Indian artwork.',
    ->  'Painting', 'West Bengal', '1935-01-01',
    ->  280000.00, '2000-07-15', 'Good'),
    ->
    -> (13, 7, 8, 'ART-013',
    ->  'Temple Bronze Idol',
    ->  'Traditional bronze idol from a South Indian temple.',
    ->  'Religious Artifact', 'Tamil Nadu', '1450-01-01',
    ->  400000.00, '1998-04-18', 'Damaged'),
    ->
    -> (14, 7, 8, 'ART-014',
    ->  'Temple Stone Panel',
    ->  'Decorative stone panel from an ancient temple.',
    ->  'Sculpture', 'Tamil Nadu', '1500-01-01',
    ->  275000.00, '1999-08-20', 'Good'),
    ->
    -> (15, 8, 8, 'ART-015',
    ->  'Traditional Wooden Doll',
    ->  'Traditional South Indian wooden craft.',
    ->  'Craft', 'Karnataka', '1880-01-01',
    ->  65000.00, '2001-05-10', 'Good'),
    ->
    -> (16, 8, 8, 'ART-016',
    ->  'Handwoven Textile',
    ->  'Traditional handwoven textile with cultural patterns.',
    ->  'Textile', 'Tamil Nadu', '1900-01-01',
    ->  95000.00, '2002-09-14', 'Needs Restoration'),
    ->
    -> (17, 9, 9, 'ART-017',
    ->  'Ancient Bronze Vessel',
    ->  'Bronze vessel discovered during archaeological research.',
    ->  'Vessel', 'Andhra Pradesh', '1100-01-01',
    ->  600000.00, '2003-01-22', 'Good'),
    ->
    -> (18, 9, 9, 'ART-018',
    ->  'Ancient Terracotta Figure',
    ->  'Terracotta figure from an archaeological excavation.',
    ->  'Terracotta', 'Telangana', '1000-01-01',
    ->  220000.00, '2004-03-18', 'Damaged'),
    ->
    -> (19, 10, 10, 'ART-019',
    ->  'Ancient Gold Coin',
    ->  'Historical gold coin from a royal period.',
    ->  'Coin', 'Hyderabad', '1650-01-01',
    ->  750000.00, '2005-06-10', 'Good'),
    ->
    -> (20, 10, 10, 'ART-020',
    ->  'Silver Royal Coin',
    ->  'Historical silver coin belonging to a royal collection.',
    ->  'Coin', 'Hyderabad', '1700-01-01',
    ->  180000.00, '2006-08-25', 'Good');
Query OK, 20 rows affected (0.01 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO EXHIBITION
    -> (exhibition_id, museum_id, exhibition_name, description, theme, start_date, end_date, status)
    -> VALUES
    -> (1, 1, 'Indian Heritage Exhibition',
    ->  'Exhibition showcasing important Indian historical artifacts.',
    ->  'Indian Heritage', '2026-01-01', '2026-06-30', 'Active'),
    ->
    -> (2, 1, 'Ancient Sculptures',
    ->  'Collection of ancient Indian sculptures.',
    ->  'Ancient Art', '2026-03-01', '2026-08-31', 'Active'),
    ->
    -> (3, 2, 'Royal Mysuru Collection',
    ->  'Artifacts from the Mysuru royal period.',
    ->  'Royal Heritage', '2026-02-01', '2026-12-31', 'Active'),
    ->
    -> (4, 2, 'Palace Treasures',
    ->  'Royal furniture and ceremonial objects.',
    ->  'Royal Treasures', '2026-04-01', '2026-09-30', 'Active'),
    ->
    -> (5, 3, 'Masters of Indian Painting',
    ->  'Exhibition of famous Indian paintings.',
    ->  'Indian Painting', '2026-01-15', '2026-05-30', 'Completed'),
    ->
    -> (6, 4, 'Temple Heritage',
    ->  'Traditional temple artifacts from South India.',
    ->  'Temple Culture', '2026-05-01', '2026-10-31', 'Active'),
    ->
    -> (7, 4, 'Traditional Crafts',
    ->  'Exhibition of traditional South Indian crafts.',
    ->  'Traditional Crafts', '2026-06-01', '2026-11-30', 'Planned'),
    ->
    -> (8, 5, 'Archaeological Discoveries',
    ->  'Important archaeological discoveries.',
    ->  'Archaeology', '2026-02-15', '2026-07-31', 'Active');
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO DISPLAY
    -> (exhibition_id, artifact_id, display_start_date, display_end_date,
    ->  display_location, display_case, display_notes)
    -> VALUES
    -> (1, 1, '2026-01-01', '2026-06-30', 'Gallery A', 'A1', 'Ancient sculpture section'),
    -> (1, 3, '2026-01-01', '2026-06-30', 'Gallery A', 'A2', 'Historical manuscripts'),
    -> (1, 9, '2026-01-01', '2026-06-30', 'Gallery B', 'B1', 'Indian paintings'),
    ->
    -> (2, 2, '2026-03-01', '2026-08-31', 'Gallery C', 'C1', 'Granite sculpture'),
    -> (2, 14, '2026-03-01', '2026-08-31', 'Gallery C', 'C2', 'Temple stone panel'),
    ->
    -> (3, 5, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R1', 'Royal furniture'),
    -> (3, 6, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R2', 'Royal ceremonial sword'),
    -> (3, 7, '2026-02-01', '2026-12-31', 'Royal Gallery', 'R3', 'Royal throne'),
    ->
    -> (4, 8, '2026-04-01', '2026-09-30', 'Palace Gallery', 'P1', 'Royal jewellery box'),
    ->
    -> (5, 10, '2026-01-15', '2026-05-30', 'Painting Gallery', 'PG1', 'Bengal School painting'),
    -> (5, 11, '2026-01-15', '2026-05-30', 'Painting Gallery', 'PG2', 'Village life painting'),
    ->
    -> (6, 13, '2026-05-01', '2026-10-31', 'Temple Gallery', 'T1', 'Bronze idol'),
    -> (6, 14, '2026-05-01', '2026-10-31', 'Temple Gallery', 'T2', 'Temple stone panel'),
    ->
    -> (8, 17, '2026-02-15', '2026-07-31', 'Archaeology Gallery', 'AR1', 'Bronze vessel'),
    -> (8, 18, '2026-02-15', '2026-07-31', 'Archaeology Gallery', 'AR2', 'Terracotta figure');
Query OK, 15 rows affected (0.01 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO ARTIFACT_MEDIA
    -> (media_id, artifact_id, staff_id, media_type, file_name, file_url, description, file_size)
    -> VALUES
    -> (1, 1, 1, 'Image', 'artifact001.jpg',
    ->  'https://museum.com/media/artifact001.jpg',
    ->  'Front view of stone sculpture.', 250000),
    ->
    -> (2, 2, 1, 'Image', 'artifact002.jpg',
    ->  'https://museum.com/media/artifact002.jpg',
    ->  'Granite temple statue image.', 280000),
    ->
    -> (3, 3, 3, 'Document', 'artifact003.pdf',
    ->  'https://museum.com/media/artifact003.pdf',
    ->  'Digital copy of royal manuscript.', 850000),
    ->
    -> (4, 5, 1, 'Image', 'artifact005.jpg',
    ->  'https://museum.com/media/artifact005.jpg',
    ->  'Royal wooden chest image.', 310000),
    ->
    -> (5, 6, 1, 'Image', 'artifact006.jpg',
    ->  'https://museum.com/media/artifact006.jpg',
    ->  'Royal sword image.', 290000),
    ->
    -> (6, 7, 7, 'Image', 'artifact007.jpg',
    ->  'https://museum.com/media/artifact007.jpg',
    ->  'Royal throne image.', 350000),
    ->
    -> (7, 9, 1, 'Image', 'artifact009.jpg',
    ->  'https://museum.com/media/artifact009.jpg',
    ->  'Mythological painting photograph.', 420000),
    ->
    -> (8, 10, 7, 'Image', 'artifact010.jpg',
    ->  'https://museum.com/media/artifact010.jpg',
    ->  'Bharat Mata painting photograph.', 450000),
    ->
```

```
    ->
    -> (9, 13, 5, 'Image', 'artifact013.jpg',
    ->  'https://museum.com/media/artifact013.jpg',
    ->  'Temple bronze idol photograph.', 300000),
    ->
    -> (10, 15, 5, 'Image', 'artifact015.jpg',
    ->  'https://museum.com/media/artifact015.jpg',
    ->  'Traditional wooden doll photograph.', 180000),
    ->
    -> (11, 17, 6, 'Image', 'artifact017.jpg',
    ->  'https://museum.com/media/artifact017.jpg',
    ->  'Ancient bronze vessel photograph.', 320000),
    ->
    -> (12, 19, 6, 'Image', 'artifact019.jpg',
    ->  'https://museum.com/media/artifact019.jpg',
    ->  'Ancient gold coin photograph.', 150000);
Query OK, 12 rows affected (0.01 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO ARTIFACT_HISTORY
    -> (history_id, artifact_id, staff_id, event_date, event_type,
    ->  location, description, source_reference)
    -> VALUES
    -> (1, 1, 1, '1986-03-12', 'Acquisition',
    ->  'National Heritage Museum',
    ->  'Artifact acquired and added to museum collection.',
    ->  'Acquisition Register 1986'),
    ->
    -> (2, 2, 1, '1987-04-15', 'Acquisition',
    ->  'National Heritage Museum',
    ->  'Granite statue added to collection.',
    ->  'Acquisition Register 1987'),
    ->
    -> (3, 3, 3, '1988-04-10', 'Acquisition',
    ->  'National Heritage Museum',
    ->  'Royal manuscript added to archives.',
    ->  'Archive Register 1988'),
    ->
    -> (4, 5, 1, '1994-02-15', 'Acquisition',
    ->  'Royal History Museum',
    ->  'Royal wooden chest acquired.',
    ->  'Royal Collection Register'),
    ->
    -> (5, 6, 1, '1995-07-11', 'Acquisition',
    ->  'Royal History Museum',
    ->  'Royal sword acquired.',
    ->  'Royal Collection Register'),
    ->
    -> (6, 7, 2, '1996-01-20', 'Acquisition',
    ->  'Royal History Museum',
    ->  'Royal throne added to collection.',
    ->  'Royal Collection Register'),
    ->
    -> (7, 9, 1, '1987-06-20', 'Acquisition',
    ->  'Indian Art Museum',
    ->  'Mythological painting acquired.',
    ->  'Painting Register'),
    ->
    -> (8, 10, 7, '1990-02-11', 'Acquisition',
    ->  'Indian Art Museum',
    ->  'Bharat Mata painting acquired.',
    ->  'Painting Register'),
    ->
    -> (9, 11, 7, '1995-03-22', 'Acquisition',
    ->  'Indian Art Museum',
    ->  'Village life painting acquired.',
    ->  'Painting Register'),
```

```
    ->
    -> (10, 13, 1, '1998-04-18', 'Acquisition',
    ->  'South Indian Heritage Museum',
    ->  'Bronze temple idol acquired.',
    ->  'Temple Artifact Register'),
    ->
    -> (11, 14, 1, '1999-08-20', 'Acquisition',
    ->  'South Indian Heritage Museum',
    ->  'Temple stone panel acquired.',
    ->  'Temple Artifact Register'),
    ->
    -> (12, 15, 6, '2001-05-10', 'Acquisition',
    ->  'South Indian Heritage Museum',
    ->  'Traditional wooden doll acquired.',
    ->  'Craft Register'),
    ->
    -> (13, 17, 6, '2003-01-22', 'Discovery',
    ->  'Ancient Civilisation Museum',
    ->  'Bronze vessel obtained through archaeological research.',
    ->  'Archaeology Report 2003'),
    ->
    -> (14, 18, 6, '2004-03-18', 'Discovery',
    ->  'Ancient Civilisation Museum',
    ->  'Terracotta figure discovered during excavation.',
    ->  'Archaeology Report 2004'),
    ->
    -> (15, 19, 6, '2005-06-10', 'Acquisition',
    ->  'Ancient Civilisation Museum',
    ->  'Ancient gold coin acquired.',
    ->  'Numismatic Register');
Query OK, 15 rows affected (0.01 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO RESTORATION_PROJECT
    -> (project_id, artifact_id, manager_id, project_name, objective,
    ->  start_date, end_date, status, cost)
    -> VALUES
    -> (1, 9, 2, 'Mythological Painting Restoration',
    ->  'Restore damaged paint layers and improve structural stability.',
    ->  '2026-01-15', '2026-05-15', 'In Progress', 75000.00),
    ->
    -> (2, 5, 2, 'Royal Wooden Chest Restoration',
    ->  'Repair structural damage and restore original surface finish.',
    ->  '2026-02-01', '2026-07-01', 'Planned', 55000.00),
    ->
    -> (3, 7, 2, 'Royal Throne Conservation',
    ->  'Restore damaged wooden structure and decorative elements.',
    ->  '2026-03-01', '2026-08-30', 'In Progress', 120000.00),
    ->
    -> (4, 13, 5, 'Bronze Idol Conservation',
    ->  'Clean and preserve the historical bronze surface.',
    ->  '2026-04-01', '2026-06-30', 'Planned', 65000.00),
    ->
    -> (5, 16, 5, 'Textile Restoration',
    ->  'Restore damaged fibers and preserve traditional patterns.',
    ->  '2026-05-01', '2026-09-30', 'Planned', 45000.00),
    ->
    -> (6, 18, 2, 'Terracotta Figure Restoration',
    ->  'Repair cracks and stabilize the terracotta artifact.',
    ->  '2026-02-15', '2026-05-30', 'Completed', 38000.00),
    ->
    -> (7, 14, 5, 'Temple Stone Panel Conservation',
    ->  'Clean and stabilize the historical stone panel.',
    ->  '2026-06-01', '2026-10-31', 'Planned', 70000.00),
    ->
    -> (8, 4, 2, 'Historical Document Preservation',
    ->  'Preserve and digitally archive the historical document.',
    ->  '2026-01-10', '2026-04-30', 'Completed', 30000.00);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO CONDITION_ASSESSMENT
    -> (assessment_id, artifact_id, staff_id, assessment_date,
    ->  conditions, damage_type, damage_description,
    ->  recommended_action, priority)
    -> VALUES
    -> (1, 9, 2, '2026-01-10','Poor', 'Surface Damage',
    ->  'Paint layer is damaged in multiple areas.','Clean surface and restore damaged paint.',
    ->  'High'),
    ->
    -> (2, 5, 2, '2026-01-20','Fair', 'Structural Damage',
    ->  'Wooden structure has cracks and weakened joints.','Repair joints and treat damaged wood.',
    ->  'Medium'),
    ->
    -> (3, 7, 2, '2026-02-01','Poor', 'Structural Damage',
    ->  'Several wooden joints are unstable.','Structural restoration required.',
    ->  'High'),
    ->
    -> (4, 13, 5, '2026-02-10','Fair', 'Surface Corrosion',
    ->  'Bronze surface has developed oxidation.','Clean and apply conservation coating.',
    ->  'Medium'),
    ->
    -> (5, 16, 5, '2026-02-15','Poor', 'Fiber Damage',
    ->  'Several textile fibers are weakened.','Textile conservation required.',
    ->  'High'),
    ->
    -> (6, 18, 2, '2026-02-20','Poor', 'Cracking',
    ->  'Terracotta figure contains multiple cracks.','Stabilize and repair cracks.',
    ->  'High'),
```

```
    ->  'High');
    ->
    -> (7, 14, 5, '2026-03-05','Fair', 'Surface Wear',
    ->  'Stone surface shows signs of weathering.','Surface cleaning and stabilization.',
    ->  'Medium'),
    ->
    -> (8, 2, 1, '2026-03-10','Good', 'Minor Wear',
    ->  'Minor surface wear observed.','Routine monitoring.',
    ->  'Low'),
    ->
    -> (9, 3, 3, '2026-03-15','Good', 'Paper Aging',
    ->  'Paper has minor signs of aging.','Climate-controlled storage.',
    ->  'Low'),
    ->
    -> (10, 6, 1, '2026-03-20','Good', 'Minor Scratches',
    ->  'Minor scratches observed on surface.','Routine conservation.',
    ->  'Low'),
    ->
    -> (11, 10, 7, '2026-03-25','Good', 'Minor Wear',
    ->  'Minor frame and surface wear.','Routine conservation.',
    ->  'Low'),
    ->
    -> (12, 15, 5, '2026-04-01','Good', 'Surface Wear',
    ->  'Minor wear due to handling.','Preventive conservation.',
    ->  'Low');
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO RESTORATION_EXPERT
    -> (expert_id, expert_name, email, phone, hours_worked)
    -> VALUES
    -> (1, 'Dr. Arjun Kumar', 'arjun@restoration.com', '9988700001', 120.00),
    ->
    -> (2, 'Priya Menon', 'priya@restoration.com', '9988700002', 85.50),
    ->
    -> (3, 'Vikram Rao', 'vikram@restoration.com', '9988700003', 60.00),
    ->
    -> (4, 'Kavya Iyer', 'kavya@restoration.com', '9988700004', 95.00),
    ->
    -> (5, 'Suresh Bhat', 'suresh@restoration.com', '9988700005', 72.50),
    ->
    -> (6, 'Neha Sharma', 'neha@restoration.com', '9988700006', 110.00),
    ->
    -> (7, 'Rohan Das', 'rohan@restoration.com', '9988700007', 65.00),
    ->
    -> (8, 'Lakshmi Rao', 'lakshmi@restoration.com', '9988700008', 88.00);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO PROJECT_EXPERT
    -> (project_id, expert_id)
    -> VALUES
    -> (1, 1),
    -> (1, 2),
    -> (1, 4),
    ->
    -> (2, 2),
    -> (2, 3),
    ->
    -> (3, 1),
    -> (3, 5),
    -> (3, 6),
    ->
    -> (4, 4),
    -> (4, 7),
    ->
    -> (5, 6),
    -> (5, 8),
    ->
    -> (6, 3),
    -> (6, 5),
    ->
    -> (7, 1);
Query OK, 15 rows affected (0.01 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO RESTORATION_MATERIAL
    -> (material_id, material_name, material_type, supplier, unit,
    ->  unit_cost, quantity_used, total_cost)
    -> VALUES
    -> (1, 'Archival Adhesive', 'Adhesive',
    ->  'Heritage Supplies', 'litre', 2500.00, 5.00, 12500.00),
    ->
    -> (2, 'Protective Varnish', 'Coating',
    ->  'Heritage Supplies', 'litre', 3500.00, 4.00, 14000.00),
    ->
    -> (3, 'Wood Treatment Oil', 'Wood Treatment',
    ->  'Conservation Materials Ltd', 'litre', 1800.00, 6.00, 10800.00),
    ->
    -> (4, 'Cleaning Solution', 'Cleaning',
    ->  'Conservation Materials Ltd', 'litre', 1200.00, 5.00, 6000.00),
    ->
    -> (5, 'Archival Paper', 'Paper',
    ->  'Museum Supplies India', 'sheet', 150.00, 100.00, 15000.00),
    ->
    -> (6, 'Protective Glass', 'Glass',
    ->  'Heritage Supplies', 'sqft', 850.00, 20.00, 17000.00),
    ->
    -> (7, 'Gold Leaf', 'Decorative',
    ->  'Traditional Materials Co', 'sheet', 450.00, 50.00, 22500.00),
    ->
    -> (8, 'Bronze Cleaner', 'Cleaning',
    ->  'Conservation Materials Ltd', 'litre', 2200.00, 4.00, 8800.00),
    ->
    -> (9, 'Textile Support Fabric', 'Textile',
    ->  'Heritage Textiles', 'metre', 900.00, 15.00, 13500.00),
    ->
    -> (10, 'Stone Consolidant', 'Stone Treatment',
    ->  'Heritage Supplies', 'litre', 3200.00, 5.00, 16000.00);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO PROJECT_MATERIAL
    -> (project_id, material_id)
    -> VALUES
    -> (1, 1),
    -> (1, 2),
    -> (1, 4),
    ->
    -> (2, 3),
    -> (2, 4),
    -> (2, 7),
    ->
    -> (3, 3),
    -> (3, 6),
    -> (3, 7),
    ->
    -> (4, 4),
    -> (4, 8),
    ->
    -> (5, 5),
    -> (5, 9),
    ->
    -> (6, 1),
    -> (6, 4),
    ->
    -> (7, 4),
    -> (7, 10),
    ->
    -> (8, 1),
    -> (8, 5),
    -> (8, 6);
Query OK, 20 rows affected (0.01 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO VISITOR
    -> (visitor_id, visitor_name, email, phone, age, nationality, registration_date)
    -> VALUES
    -> (1, 'Aarav Singh', 'aarav@gmail.com', '9000000001', 24, 'Indian', '2026-01-05'),
    ->
    -> (2, 'Ananya Rao', 'ananya@gmail.com', '9000000002', 21, 'Indian', '2026-01-10'),
    ->
    -> (3, 'John Smith', 'john@gmail.com', '9000000003', 35, 'British', '2026-02-15'),
    ->
    -> (4, 'Sophia Lee', 'sophia@gmail.com', '9000000004', 28, 'Singaporean', '2026-03-01'),
    ->
    -> (5, 'Rahul Verma', 'rahulv@gmail.com', '9000000005', 30, 'Indian', '2026-03-05'),
    ->
    -> (6, 'Sneha Kapoor', 'sneha@gmail.com', '9000000006', 22, 'Indian', '2026-03-12'),
    ->
    -> (7, 'David Wilson', 'david@gmail.com', '9000000007', 42, 'British', '2026-03-20'),
    ->
    -> (8, 'Emily Brown', 'emily@gmail.com', '9000000008', 31, 'American', '2026-03-25'),
    ->
    -> (9, 'Vikram Shetty', 'vikram@gmail.com', '9000000009', 26, 'Indian', '2026-04-01'),
    ->
    -> (10, 'Priya Nair', 'priya@gmail.com', '9000000010', 23, 'Indian', '2026-04-05'),
    ->
    -> (11, 'Michael Chen', 'michael@gmail.com', '9000000011', 38, 'Singaporean', '2026-04-10'),
    ->
    -> (12, 'Kavya Menon', 'kavya@gmail.com', '9000000012', 27, 'Indian', '2026-04-15'),
    ->
    -> (13, 'Daniel Garcia', 'daniel@gmail.com', '9000000013', 34, 'Spanish', '2026-04-20'),
    ->
    -> (14, 'Meera Iyer', 'meera.visitor@gmail.com', '9000000014', 29, 'Indian', '2026-04-25'),
    ->
    -> (15, 'Lucas Martin', 'lucas@gmail.com', '9000000015', 40, 'French', '2026-05-01');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO VISIT_RECORD
    -> (visit_id, visitor_id, museum_id, visit_date,
    ->  entry_time, exit_time, visit_type, ticket_type)
    -> VALUES
    -> (1, 1, 1, '2026-02-10', '10:00:00', '13:00:00', 'Individual', 'General'),
    ->
    -> (2, 2, 1, '2026-02-12', '11:00:00', '14:00:00', 'Individual', 'Student'),
    ->
    -> (3, 3, 2, '2026-03-05', '09:30:00', '12:30:00', 'Tourist', 'General'),
    ->
    -> (4, 4, 2, '2026-03-10', '14:00:00', '17:00:00', 'Tourist', 'General'),
    ->
    -> (5, 5, 1, '2026-03-15', '10:30:00', '13:30:00', 'Individual', 'General'),
    ->
    -> (6, 6, 3, '2026-03-18', '11:00:00', '15:00:00', 'Student', 'Student'),
    ->
    -> (7, 7, 3, '2026-03-20', '09:00:00', '12:00:00', 'Tourist', 'General'),
    ->
    -> (8, 8, 4, '2026-03-22', '10:00:00', '13:00:00', 'Tourist', 'General'),
    ->
    -> (9, 9, 4, '2026-03-25', '12:00:00', '15:00:00', 'Individual', 'Student'),
    ->
    -> (10, 10, 5, '2026-03-28', '09:30:00', '12:30:00', 'Tourist', 'General'),
    ->
    -> (11, 11, 5, '2026-04-01', '10:00:00', '14:00:00', 'Tourist', 'General'),
    ->
    -> (12, 12, 1, '2026-04-03', '11:00:00', '13:00:00', 'Individual', 'Student'),
    ->
    -> (13, 13, 2, '2026-04-05', '09:00:00', '12:00:00', 'Tourist', 'General'),
    ->
    -> (14, 14, 3, '2026-04-08', '13:00:00', '16:00:00', 'Individual', 'General'),
    ->
    -> (15, 15, 4, '2026-04-10', '10:30:00', '14:00:00', 'Tourist', 'General'),
    ->
    -> (16, 1, 5, '2026-04-12', '09:00:00', '12:00:00', 'Individual', 'General'),
    ->
    -> (17, 2, 2, '2026-04-15', '10:00:00', '13:00:00', 'Individual', 'Student'),
    ->
    -> (18, 3, 1, '2026-04-18', '11:00:00', '14:30:00', 'Tourist', 'General'),
    ->
    -> (19, 4, 3, '2026-04-20', '09:30:00', '12:30:00', 'Tourist', 'General'),
    ->
```

```
    ->
    -> (20, 5, 4, '2026-04-22', '10:00:00', '13:00:00', 'Individual', 'General'),
    ->
    -> (21, 6, 5, '2026-04-25', '12:00:00', '15:00:00', 'Student', 'Student'),
    ->
    -> (22, 7, 1, '2026-04-27', '09:00:00', '11:30:00', 'Tourist', 'General'),
    ->
    -> (23, 8, 2, '2026-04-29', '10:00:00', '13:00:00', 'Individual', 'General'),
    ->
    -> (24, 9, 3, '2026-05-01', '11:00:00', '14:00:00', 'Student', 'Student'),
    ->
    -> (25, 10, 5, '2026-05-03', '09:30:00', '12:30:00', 'Tourist', 'General');
Query OK, 25 rows affected (0.01 sec)
Records: 25  Duplicates: 0  Warnings: 0
```

```
mysql> UPDATE ARTIFACT
    -> SET estimated_value = 300000.00
    -> WHERE artifact_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> UPDATE ARTIFACT
    -> SET status = 'Needs Restoration'
    -> WHERE artifact_id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> DELETE FROM VISITOR
    -> WHERE visitor_id = 15;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select *from artifact;
+-------------+---------------+------------+------------------+-----------------------+-----------------------------------------------------+---------------------+--------------+---------------+-----------------+-----------------+-----------------------+
| artifact_id | collection_id | creator_id | accession_number | artifact_name         | description                                         | artifact_type       | origin_place | creation_date | estimated_value | acquisition_date | status                |
+-------------+---------------+------------+------------------+-----------------------+-----------------------------------------------------+---------------------+--------------+---------------+-----------------+-----------------+-----------------------+
|           1 |             1 |          2 | ART-001          | Ancient Stone Sculpture | Stone sculpture depicting an ancient Indian deity.  | Sculpture           | Hampi        | 1250-01-01    |       300000.00 | 1986-03-12      | Good                  |
|           2 |             1 |          2 | ART-002          | Granite Temple Statue | Carved granite statue from a historic temple.       | Sculpture           | Hampi        | 1300-01-01    |       320000.00 | 1987-04-15      | Needs Restoration     |
|           3 |             2 |          4 | ART-003          | Royal Manuscript      | Handwritten manuscript containing historical records. | Manuscript          | Mysuru       | 1650-01-01    |       150000.00 | 1988-04-10      | Good                  |
|           4 |             2 |          4 | ART-004          | Historical Document   | Historical administrative document.                 | Document            | Karnataka    | 1700-01-01    |        80000.00 | 1990-08-19      | Good                  |
|           5 |             3 |          3 | ART-005          | Royal Wooden Chest    | Decorated wooden chest used by a historical royal family. | Furniture           | Mysuru       | 1700-01-01    |       180000.00 | 1994-02-15      | Damaged               |
|           6 |             3 |          3 | ART-006          | Royal Sword           | Ceremonial sword belonging to a royal collection.   | Weapon              | Mysuru       | 1775-01-01    |       300000.00 | 1995-07-11      | Good                  |
|           7 |             4 |          3 | ART-007          | Royal Throne          | Decorated wooden throne used for ceremonial purposes. | Furniture           | Mysuru       | 1800-01-01    |       550000.00 | 1996-01-20      | Needs Restoration     |
|           8 |             4 |          3 | ART-008          | Royal Jewellery Box   | Ornamented jewellery box from the royal collection. | Furniture           | Mysuru       | 1820-01-01    |       210000.00 | 1997-05-12      | Good                  |
|           9 |             5 |          1 | ART-009          | Mythological Painting | Painting depicting an Indian mythological scene.    | Painting            | Mysuru       | 1890-01-01    |       450000.00 | 1987-06-20      | Needs Restoration     |
|          10 |             5 |          5 | ART-010          | Bharat Mata Painting  | Traditional Indian painting representing Bharat Mata. | Painting            | Kolkata      | 1905-01-01    |       580000.00 | 1999-02-11      | Good                  |
|          11 |             5 |          6 | ART-011          | Village Life Painting | Painting depicting rural Indian life.               | Painting            | Kolkata      | 1920-01-01    |       350000.00 | 1995-03-22      | Good                  |
|          12 |             6 |          7 | ART-012          | Folk Art Composition  | Folk-inspired modern Indian artwork.                | Painting            | West Bengal  | 1935-01-01    |       280000.00 | 2000-07-15      | Good                  |
|          13 |             7 |          8 | ART-013          | Temple Bronze Idol    | Traditional bronze idol from a South Indian temple. | Religious Artifact  | Tamil Nadu   | 1450-01-01    |       480000.00 | 1998-04-18      | Damaged               |
|          14 |             7 |          8 | ART-014          | Temple Stone Panel    | Decorative stone panel from an ancient temple.      | Sculpture           | Tamil Nadu   | 1500-01-01    |       275000.00 | 1999-08-20      | Good                  |
|          15 |             8 |          8 | ART-015          | Traditional Wooden Doll | Traditional South Indian wooden craft.              | Craft               | Karnataka    | 1880-01-01    |        65000.00 | 2001-05-10      | Good                  |
|          16 |             8 |          8 | ART-016          | Handwoven Textile     | Traditional handwoven textile with cultural patterns. | Textile             | Tamil Nadu   | 1900-01-01    |        95000.00 | 2002-09-14      | Needs Restoration     |
|          17 |             9 |          9 | ART-017          | Ancient Bronze Vessel | Bronze vessel discovered during archaeological research. | Vessel              | Andhra Pradesh | 1100-01-01    |       600000.00 | 2003-01-22      | Good                  |
|          18 |             9 |          9 | ART-018          | Ancient Terracotta Figure | Terracotta figure from an archaeological excavation. | Terracotta          | Telangana    | 1000-01-01    |       220000.00 | 2004-03-18      | Damaged               |
|          19 |            10 |         10 | ART-019          | Ancient Gold Coin     | Historical gold coin from a royal period.           | Coin                | Hyderabad    | 1650-01-01    |       750000.00 | 2005-06-10      | Good                  |
|          20 |            10 |         10 | ART-020          | Silver Royal Coin     | Historical silver coin belonging to a royal collection. | Coin                | Hyderabad    | 1700-01-01    |       180000.00 | 2006-08-25      | Good                  |
+-------------+---------------+------------+------------------+-----------------------+-----------------------------------------------------+---------------------+--------------+---------------+-----------------+-----------------+-----------------------+
20 rows in set (0.00 sec)
```

```
mysql> select *from artifact_history;
+------------+-------------+----------+------------+------------+-----------------+-----------------------------------------------------+-----------------------------+
| history_id | artifact_id | staff_id | event_date | event_type | location        | description                                         | source_reference            |
+------------+-------------+----------+------------+------------+-----------------+-----------------------------------------------------+-----------------------------+
|          1 |           1 |        1 | 1986-03-12 | Acquisition | National Heritage Museum | Artifact acquired and added to museum collection.   | Acquisition Register 1986   |
|          2 |           2 |        1 | 1987-04-15 | Acquisition | National Heritage Museum | Granite statue added to collection.                 | Acquisition Register 1987   |
|          3 |           3 |        3 | 1988-04-10 | Acquisition | National Heritage Museum | Royal manuscript added to archives.                 | Archive Register 1988       |
|          4 |           5 |        1 | 1994-02-15 | Acquisition | Royal History Museum | Royal wooden chest acquired.                        | Royal Collection Register   |
|          5 |           6 |        1 | 1995-07-11 | Acquisition | Royal History Museum | Royal sword acquired.                               | Royal Collection Register   |
|          6 |           7 |        2 | 1996-01-20 | Acquisition | Royal History Museum | Royal throne added to collection.                   | Royal Collection Register   |
|          7 |           9 |        1 | 1987-06-20 | Acquisition | Indian Art Museum | Mythological painting acquired.                     | Painting Register           |
|          8 |          10 |        7 | 1990-02-11 | Acquisition | Indian Art Museum | Bharat Mata painting acquired.                      | Painting Register           |
|          9 |          11 |        7 | 1995-03-22 | Acquisition | Indian Art Museum | Village life painting acquired.                     | Painting Register           |
|         10 |          13 |        1 | 1998-04-18 | Acquisition | South Indian Heritage Museum | Bronze temple idol acquired.                        | Temple Artifact Register    |
|         11 |          14 |        1 | 1999-08-20 | Acquisition | South Indian Heritage Museum | Temple stone panel acquired.                        | Temple Artifact Register    |
|         12 |          15 |        6 | 2001-05-10 | Acquisition | South Indian Heritage Museum | Traditional wooden doll acquired.                   | Craft Register              |
|         13 |          17 |        6 | 2003-01-22 | Discovery  | Ancient Civilisation Museum | Bronze vessel obtained through archaeological research. | Archaeology Report 2003     |
|         14 |          18 |        6 | 2004-03-18 | Discovery  | Ancient Civilisation Museum | Terracotta figure discovered during excavation.     | Archaeology Report 2004     |
|         15 |          19 |        6 | 2005-06-10 | Acquisition | Ancient Civilisation Museum | Ancient gold coin acquired.                         | Numismatic Register         |
+------------+-------------+----------+------------+------------+-----------------+-----------------------------------------------------+-----------------------------+
15 rows in set (0.00 sec)
```

```
mysql> select *from artifact_media;
+----------+-------------+----------+------------+---------------+-----------------------------------------------+-----------------------------------------------+---------------------+-----------+
| media_id | artifact_id | staff_id | media_type | file_name     | file_url                                      | description                                   | uploaded_on         | file_size |
+----------+-------------+----------+------------+---------------+-----------------------------------------------+-----------------------------------------------+---------------------+-----------+
|        1 |           1 |        1 | Image      | artifact001.jpg | https://museum.com/media/artifact001.jpg      | Front view of stone sculpture.                | 2026-10-03 17:56:34 |    250000 |
|        2 |           2 |        1 | Image      | artifact002.jpg | https://museum.com/media/artifact002.jpg      | Granite temple statue image.                  | 2026-10-03 17:56:34 |    280000 |
|        3 |           3 |        3 | Document   | artifact003.pdf | https://museum.com/media/artifact003.pdf      | Digital copy of royal manuscript.             | 2026-10-03 17:56:34 |    850000 |
|        4 |           5 |        1 | Image      | artifact005.jpg | https://museum.com/media/artifact005.jpg      | Royal wooden chest image.                     | 2026-10-03 17:56:34 |    310000 |
|        5 |           6 |        1 | Image      | artifact006.jpg | https://museum.com/media/artifact006.jpg      | Royal sword image.                            | 2026-10-03 17:56:34 |    290000 |
|        6 |           7 |        7 | Image      | artifact007.jpg | https://museum.com/media/artifact007.jpg      | Royal throne image.                           | 2026-10-03 17:56:34 |    350000 |
|        7 |           9 |        1 | Image      | artifact009.jpg | https://museum.com/media/artifact009.jpg      | Mythological painting photograph.             | 2026-10-03 17:56:34 |    420000 |
|        8 |          10 |        7 | Image      | artifact010.jpg | https://museum.com/media/artifact010.jpg      | Bharat Mata painting photograph.              | 2026-10-03 17:56:34 |    450000 |
|        9 |          13 |        5 | Image      | artifact013.jpg | https://museum.com/media/artifact013.jpg      | Temple bronze idol photograph.                | 2026-10-03 17:56:34 |    300000 |
|       10 |          15 |        5 | Image      | artifact015.jpg | https://museum.com/media/artifact015.jpg      | Traditional wooden doll photograph.           | 2026-10-03 17:56:34 |    180000 |
|       11 |          17 |        6 | Image      | artifact017.jpg | https://museum.com/media/artifact017.jpg      | Ancient bronze vessel photograph.             | 2026-10-03 17:56:34 |    320000 |
|       12 |          19 |        6 | Image      | artifact019.jpg | https://museum.com/media/artifact019.jpg      | Ancient gold coin photograph.                 | 2026-10-03 17:56:34 |    150000 |
+----------+-------------+----------+------------+---------------+-----------------------------------------------+-----------------------------------------------+---------------------+-----------+
12 rows in set (0.00 sec)
```

```
mysql> select *from collection;
```

collection_id	museum_id	collection_name	description	historical_period	collection_type	date_created
1	1	Ancient Indian Art	Collection of ancient Indian sculptures and artwork.	Ancient India	Sculpture	1986-01-10
2	1	Medieval Manuscripts	Collection of historical manuscripts and documents.	Medieval Period	Manuscript	1987-05-15
3	2	Royal Heritage	Artifacts belonging to historical royal families.	Medieval India	Royal Artifacts	1993-03-20
4	2	Mysuru Palace Collection	Objects associated with Mysuru royal history.	18th-20th Century	Royal Collection	1994-04-15
5	3	Indian Paintings	Collection of traditional Indian paintings.	19th-20th Century	Painting	1979-02-12
6	3	Modern Art	Collection of modern Indian artworks.	20th-21st Century	Modern Art	1985-07-21
7	4	Temple Artifacts	Artifacts collected from South Indian temples.	Ancient-Medieval	Religious Art	1996-03-10
8	4	Traditional Crafts	Traditional crafts and handmade cultural objects.	19th-20th Century	Craft	1997-08-18
9	5	Archaeological Collection	Artifacts discovered through archaeological research.	Ancient Period	Archaeological	2002-01-20
10	5	Coins and Currency	Historical coins and currency from different periods.	Ancient-Modern	Numismatics	2003-06-11

```
10 rows in set (0.00 sec)
```

```
mysql> select *from condition_assessment;
```

assessment_id	artifact_id	staff_id	assessment_date	conditions	damage_type	damage_description	recommended_action	priority
1	9	2	2026-01-10	Poor	Surface Damage	Paint layer is damaged in multiple areas.	Clean surface and restore damaged paint.	High
2	5	2	2026-01-20	Fair	Structural Damage	Wooden structure has cracks and weakened joints.	Repair joints and treat damaged wood.	Medium
3	7	2	2026-02-01	Poor	Structural Damage	Several wooden joints are unstable.	Structural restoration required.	High
4	13	5	2026-02-10	Fair	Surface Corrosion	Bronze surface has developed oxidation.	Clean and apply conservation coating.	Medium
5	16	5	2026-02-15	Poor	Fiber Damage	Several textile fibers are weakened.	Textile conservation required.	High
6	18	2	2026-02-20	Poor	Cracking	Terracotta figure contains multiple cracks.	Stabilize and repair cracks.	High
7	14	5	2026-03-05	Fair	Surface Wear	Stone surface shows signs of weathering.	Surface cleaning and stabilization.	Medium
8	2	1	2026-03-10	Good	Minor Wear	Minor surface wear observed.	Routine monitoring.	Low
9	3	3	2026-03-15	Good	Paper Aging	Paper has minor signs of aging.	Climate-controlled storage.	Low
10	6	1	2026-03-20	Good	Minor Scratches	Minor scratches observed on surface.	Routine conservation.	Low
11	10	7	2026-03-25	Good	Minor Wear	Minor frame and surface wear.	Routine conservation.	Low
12	15	5	2026-04-01	Good	Surface Wear	Minor wear due to handling.	Preventive conservation.	Low

```
12 rows in set (0.00 sec)
```

```
mysql> select *from creator;
```

creator_id	creator_name	birth_year	nationality	biography
1	Ravi Varma	1848	Indian	Indian painter known for paintings based on Indian mythology.
2	Unknown Ancient Sculptor	1200	Indian	Ancient sculptor whose identity remains unknown.
3	Mysuru Royal Artisan	1750	Indian	Traditional artisan associated with the Mysuru royal court.
4	Unknown Manuscript Artist	1600	Indian	Artist responsible for illustrating historical manuscripts.
5	Abanindranath Tagore	1871	Indian	Indian painter and one of the founders of the Bengal School of Art.
6	Nandalal Bose	1882	Indian	Indian painter associated with the Bengal School of Art.
7	Jamini Roy	1887	Indian	Indian painter known for folk-inspired artistic styles.
8	Unknown Temple Artisan	1400	Indian	Traditional artisan responsible for temple artwork.
9	Ancient Metalworker	1100	Indian	Ancient artisan specializing in metal objects.
10	Royal Coin Maker	1650	Indian	Artisan associated with historical royal coin production.

```
10 rows in set (0.00 sec)
```

```
mysql> select *from display;
```

exhibition_id	artifact_id	display_start_date	display_end_date	display_location	display_case	display_notes
1	1	2026-01-01	2026-06-30	Gallery A	A1	Ancient sculpture section
1	3	2026-01-01	2026-06-30	Gallery A	A2	Historical manuscripts
1	9	2026-01-01	2026-06-30	Gallery B	B1	Indian paintings
2	2	2026-03-01	2026-08-31	Gallery C	C1	Granite sculpture
2	14	2026-03-01	2026-08-31	Gallery C	C2	Temple stone panel
3	5	2026-02-01	2026-12-31	Royal Gallery	R1	Royal furniture
3	6	2026-02-01	2026-12-31	Royal Gallery	R2	Royal ceremonial sword
3	7	2026-02-01	2026-12-31	Royal Gallery	R3	Royal throne
4	8	2026-04-01	2026-09-30	Palace Gallery	P1	Royal jewellery box
5	10	2026-01-15	2026-05-30	Painting Gallery	PG1	Bengal School painting
5	11	2026-01-15	2026-05-30	Painting Gallery	PG2	Village life painting
6	13	2026-05-01	2026-10-31	Temple Gallery	T1	Bronze idol
6	14	2026-05-01	2026-10-31	Temple Gallery	T2	Temple stone panel
8	17	2026-02-15	2026-07-31	Archaeology Gallery	AR1	Bronze vessel
8	18	2026-02-15	2026-07-31	Archaeology Gallery	AR2	Terracotta figure

```
15 rows in set (0.00 sec)
```

```
mysql> select *from exhibition;
```

exhibition_id	museum_id	exhibition_name	description	theme	start_date	end_date	status
1	1	Indian Heritage Exhibition	Exhibition showcasing important Indian historical artifacts.	Indian Heritage	2026-01-01	2026-06-30	Active
2	1	Ancient Sculptures	Collection of ancient Indian sculptures.	Ancient Art	2026-03-01	2026-08-31	Active
3	2	Royal Mysuru Collection	Artifacts from the Mysuru royal period.	Royal Heritage	2026-02-01	2026-12-31	Active
4	2	Palace Treasures	Royal furniture and ceremonial objects.	Royal Treasures	2026-04-01	2026-09-30	Active
5	3	Masters of Indian Painting	Exhibition of famous Indian paintings.	Indian Painting	2026-01-15	2026-05-30	Completed
6	4	Temple Heritage	Traditional temple artifacts from South India.	Temple Culture	2026-05-01	2026-10-31	Active
7	4	Traditional Crafts	Exhibition of traditional South Indian crafts.	Traditional Crafts	2026-06-01	2026-11-30	Planned
8	5	Archaeological Discoveries	Important archaeological discoveries.	Archaeology	2026-02-15	2026-07-31	Active

```
8 rows in set (0.00 sec)
```

```
mysql> select *from museum;
```

museum_id	museum_name	description	address	city	state	country	contact_email	contact_phone	established_year
1	National Heritage Museum	Museum preserving Indian historical and cultural artifacts.	MG Road	Bengaluru	Karnataka	India	info@nationalheritage.com	08040001001	1985
2	Royal History Museum	Museum dedicated to royal history and traditional artifacts.	Palace Road	Mysuru	Karnataka	India	info@royalhistory.com	0821H000102	1992
3	Indian Art Museum	Museum showcasing traditional and modern Indian art.	Museum Road	New Delhi	Delhi	India	info@indianartmuseum.com	01140001003	1978
4	South Indian Heritage Museum	Museum preserving cultural heritage from South India.	Temple Street	Chennai	Tamil Nadu	India	info@sihmuseum.com	04440001004	1995
5	Ancient Civilisation Museum	Museum containing archaeological and ancient civilisation artifacts.	Heritage Avenue	Hyderabad	Telangana	India	info@ancientcivilisation.com	04040001005	2001

```
5 rows in set (0.00 sec)
```

```
mysql> select *from project_expert;
+------------+-----------+
| project_id | expert_id |
+------------+-----------+
|          1 |         1 |
|          1 |         2 |
|          1 |         4 |
|          2 |         2 |
|          2 |         3 |
|          3 |         1 |
|          3 |         5 |
|          3 |         6 |
|          4 |         4 |
|          4 |         7 |
|          5 |         6 |
|          5 |         8 |
|          6 |         3 |
|          6 |         5 |
|          7 |         1 |
+------------+-----------+
15 rows in set (0.00 sec)
```

```
mysql> select *from project_material;
+------------+-------------+
| project_id | material_id |
+------------+-------------+
|          1 |           1 |
|          1 |           2 |
|          1 |           4 |
|          2 |           3 |
|          2 |           4 |
|          2 |           7 |
|          3 |           3 |
|          3 |           6 |
|          3 |           7 |
|          4 |           4 |
|          4 |           8 |
|          5 |           5 |
|          5 |           9 |
|          6 |           1 |
|          6 |           4 |
|          7 |           4 |
|          7 |          10 |
|          8 |           1 |
|          8 |           5 |
|          8 |           6 |
+------------+-------------+
20 rows in set (0.00 sec)
```

```
mysql> select *from restoration_expert;
+------------+-----------------+---------------------------+-------------+--------------+
| expert_id | expert_name     | email                     | phone       | hours_worked |
+------------+-----------------+---------------------------+-------------+--------------+
|          1 | Dr. Arjun Kumar | arjun@restoration.com     | 9988700001 |       120.00 |
|          2 | Priya Menon     | priya@restoration.com     | 9988700002 |        85.50 |
|          3 | Vikram Rao      | vikram@restoration.com    | 9988700003 |        60.00 |
|          4 | Kavya Iyer      | kavya@restoration.com     | 9988700004 |        95.00 |
|          5 | Suresh Bhat     | suresh@restoration.com    | 9988700005 |        72.50 |
|          6 | Neha Sharma     | neha@restoration.com      | 9988700006 |       110.00 |
|          7 | Rohan Das       | rohan@restoration.com     | 9988700007 |        65.00 |
|          8 | Lakshmi Rao     | lakshmi@restoration.com   | 9988700008 |        88.00 |
+------------+-----------------+---------------------------+-------------+--------------+
8 rows in set (0.00 sec)
```

```
mysql> select *from restoration_material;
+-------------+-------------------------+----------------+---------------------------+-------+-----------+---------------+------------+
| material_id | material_name           | material_type  | supplier                  | unit  | unit_cost | quantity_used | total_cost |
+-------------+-------------------------+----------------+---------------------------+-------+-----------+---------------+------------+
|           1 | Archival Adhesive       | Adhesive       | Heritage Supplies         | litre |   2500.00 |          5.00 |   12500.00 |
|           2 | Protective Varnish      | Coating        | Heritage Supplies         | litre |   3500.00 |          4.00 |   14000.00 |
|           3 | Wood Treatment Oil      | Wood Treatment | Conservation Materials Ltd | litre |   1800.00 |          6.00 |   10800.00 |
|           4 | Cleaning Solution       | Cleaning       | Conservation Materials Ltd | litre |   1200.00 |          5.00 |    6000.00 |
|           5 | Archival Paper          | Paper          | Museum Supplies India     | sheet |    150.00 |        100.00 |   15000.00 |
|           6 | Protective Glass        | Glass          | Heritage Supplies         | sqft  |    850.00 |         20.00 |   17000.00 |
|           7 | Gold Leaf               | Decorative     | Traditional Materials Co  | sheet |    450.00 |         50.00 |   22500.00 |
|           8 | Bronze Cleaner          | Cleaning       | Conservation Materials Ltd | litre |   2200.00 |          4.00 |    8800.00 |
|           9 | Textile Support Fabric  | Textile        | Heritage Textiles         | metre |    900.00 |         15.00 |   13500.00 |
|          10 | Stone Consolidant       | Stone Treatment | Heritage Supplies         | litre |   3200.00 |          5.00 |   16000.00 |
+-------------+-------------------------+----------------+---------------------------+-------+-----------+---------------+------------+
10 rows in set (0.00 sec)
```

```
mysql> select *from restoration_project;
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------------+------------+------------+-------------+-----------+
| project_id | artifact_id | manager_id | project_name                  | objective                                                       | start_date | end_date   | status      | cost      |
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------------+------------+------------+-------------+-----------+
|          1 |           9 |          2 | Mythological Painting Restoration | Restore damaged paint layers and improve structural stability. | 2026-01-15 | 2026-05-15 | In Progress |  75000.00 |
|          2 |           5 |          2 | Royal Wooden Chest Restoration | Repair structural damage and restore original surface finish.  | 2026-02-01 | 2026-07-01 | Planned     |  55000.00 |
|          3 |           7 |          2 | Royal Throne Conservation     | Restore damaged wooden structure and decorative elements.       | 2026-03-01 | 2026-08-30 | In Progress | 120000.00 |
|          4 |          13 |          5 | Bronze Idol Conservation      | Clean and preserve the historical bronze surface.               | 2026-04-01 | 2026-06-30 | Planned     |  65000.00 |
|          5 |          16 |          5 | Textile Restoration           | Restore damaged fibers and preserve traditional patterns.       | 2026-05-01 | 2026-09-30 | Planned     |  45000.00 |
|          6 |          18 |          2 | Terracotta Figure Restoration | Repair cracks and stabilize the terracotta artifact.            | 2026-02-15 | 2026-05-30 | Completed   |  38000.00 |
|          7 |          14 |          5 | Temple Stone Panel Conservation | Clean and stabilize the historical stone panel.               | 2026-06-01 | 2026-10-31 | Planned     |  70000.00 |
|          8 |           4 |          2 | Historical Document Preservation | Preserve and digitally archive the historical document.       | 2026-01-10 | 2026-04-30 | Completed   |  30000.00 |
+------------+-------------+------------+-------------------------------+-----------------------------------------------------------------+------------+------------+-------------+-----------+
8 rows in set (0.00 sec)
```

```
mysql> select *from staff
    -> ;
+----------+--------------+-------------------+------------+---------------------+----------------+-------------+---------------+--------------+
| staff_id | staff_name   | email             | phone      | designation         | department     | username    | password_hash | staff_status |
+----------+--------------+-------------------+------------+---------------------+----------------+-------------+---------------+--------------+
|        1 | Anita Rao    | anita@museum.com  | 9876500001 | Curator             | Collections    | anita.rao   | hash_anita123 | Active       |
|        2 | Rahul Sharma | rahul@museum.com  | 9876500002 | Restoration Manager | Restoration    | rahul.sharma | hash_rahul123 | Active       |
|        3 | Meera Nair   | meera@museum.com  | 9876500003 | Archivist           | Archives       | meera.nair  | hash_meera123 | Active       |
|        4 | Kiran Kumar  | kiran@museum.com  | 9876500004 | Museum Director     | Administration | kiran.kumar | hash_kiran123 | Active       |
|        5 | Priya Menon  | priya@museum.com  | 9876500005 | Conservator         | Restoration    | priya.menon | hash_priya123 | Active       |
|        6 | Vivek Rao    | vivek@museum.com  | 9876500006 | Researcher          | Research       | vivek.rao   | hash_vivek123 | Active       |
|        7 | Sneha Iyer   | sneha@museum.com  | 9876500007 | Curator             | Exhibitions    | sneha.iyer  | hash_sneha123 | Active       |
|        8 | Arjun Das    | arjun@museum.com  | 9876500008 | Security Officer    | Security       | arjun.das   | hash_arjun123 | Active       |
|        9 | Divya Shah   | divya@museum.com  | 9876500009 | Archivist           | Archives       | divya.shah  | hash_divya123 | Active       |
|       10 | Manoj Kumar  | manoj@museum.com  | 9876500010 | Technician          | Digital Archives | manoj.kumar | hash_manoj123 | Active       |
+----------+--------------+-------------------+------------+---------------------+----------------+-------------+---------------+--------------+
10 rows in set (0.00 sec)
```

```
mysql> select *from visit_record;
+----------+------------+-----------+------------+------------+------------+------------+-------------+
| visit_id | visitor_id | museum_id | visit_date | entry_time | exit_time  | visit_type | ticket_type |
+----------+------------+-----------+------------+------------+------------+------------+-------------+
|        1 |          1 |         1 | 2026-02-10 | 10:00:00   | 13:00:00   | Individual | General     |
|        2 |          2 |         1 | 2026-02-12 | 11:00:00   | 14:00:00   | Individual | Student     |
|        3 |          3 |         2 | 2026-03-05 | 09:30:00   | 12:30:00   | Tourist    | General     |
|        4 |          4 |         2 | 2026-03-10 | 14:00:00   | 17:00:00   | Tourist    | General     |
|        5 |          5 |         1 | 2026-03-15 | 10:30:00   | 13:30:00   | Individual | General     |
|        6 |          6 |         3 | 2026-03-18 | 11:00:00   | 15:00:00   | Student    | Student     |
|        7 |          7 |         3 | 2026-03-20 | 09:00:00   | 12:00:00   | Tourist    | General     |
|        8 |          8 |         4 | 2026-03-22 | 10:00:00   | 13:00:00   | Tourist    | General     |
|        9 |          9 |         4 | 2026-03-25 | 12:00:00   | 15:00:00   | Individual | Student     |
|       10 |         10 |         5 | 2026-03-28 | 09:30:00   | 12:30:00   | Tourist    | General     |
|       11 |         11 |         5 | 2026-04-01 | 10:00:00   | 14:00:00   | Tourist    | General     |
|       12 |         12 |         1 | 2026-04-03 | 11:00:00   | 13:00:00   | Individual | Student     |
|       13 |         13 |         2 | 2026-04-05 | 09:00:00   | 12:00:00   | Tourist    | General     |
|       14 |         14 |         3 | 2026-04-08 | 13:00:00   | 16:00:00   | Individual | General     |
|       16 |          1 |         5 | 2026-04-12 | 09:00:00   | 12:00:00   | Individual | General     |
|       17 |          2 |         2 | 2026-04-15 | 10:00:00   | 13:00:00   | Individual | Student     |
|       18 |          3 |         1 | 2026-04-18 | 11:00:00   | 14:30:00   | Tourist    | General     |
|       19 |          4 |         3 | 2026-04-20 | 09:30:00   | 12:30:00   | Tourist    | General     |
|       20 |          5 |         4 | 2026-04-22 | 10:00:00   | 13:00:00   | Individual | General     |
|       21 |          6 |         5 | 2026-04-25 | 12:00:00   | 15:00:00   | Student    | Student     |
|       22 |          7 |         1 | 2026-04-27 | 09:00:00   | 11:30:00   | Tourist    | General     |
|       23 |          8 |         2 | 2026-04-29 | 10:00:00   | 13:00:00   | Individual | General     |
|       24 |          9 |         3 | 2026-05-01 | 11:00:00   | 14:00:00   | Student    | Student     |
|       25 |         10 |         5 | 2026-05-03 | 09:30:00   | 12:30:00   | Tourist    | General     |
+----------+------------+-----------+------------+------------+------------+------------+-------------+
24 rows in set (0.00 sec)
```

```
mysql> select *from visitor;
+------------+--------------+---------------------+------------+------+-------------+-------------------+
| visitor_id | visitor_name | email               | phone      | age  | nationality | registration_date |
+------------+--------------+---------------------+------------+------+-------------+-------------------+
|          1 | Aarav Singh  | aarav@gmail.com     | 9000000001 |   24 | Indian      | 2026-01-05        |
|          2 | Ananya Rao   | ananya@gmail.com    | 9000000002 |   21 | Indian      | 2026-01-10        |
|          3 | John Smith   | john@gmail.com      | 9000000003 |   35 | British     | 2026-02-15        |
|          4 | Sophia Lee   | sophia@gmail.com    | 9000000004 |   28 | Singaporean | 2026-03-01        |
|          5 | Rahul Verma  | rahulv@gmail.com    | 9000000005 |   30 | Indian      | 2026-03-05        |
|          6 | Sneha Kapoor | sneha@gmail.com     | 9000000006 |   22 | Indian      | 2026-03-12        |
|          7 | David Wilson | david@gmail.com     | 9000000007 |   42 | British     | 2026-03-20        |
|          8 | Emily Brown  | emily@gmail.com     | 9000000008 |   31 | American    | 2026-03-25        |
|          9 | Vikram Shetty | vikram@gmail.com   | 9000000009 |   26 | Indian      | 2026-04-01        |
|         10 | Priya Nair   | priya@gmail.com     | 9000000010 |   23 | Indian      | 2026-04-05        |
|         11 | Michael Chen | michael@gmail.com   | 9000000011 |   38 | Singaporean | 2026-04-10        |
|         12 | Kavya Menon  | kavya@gmail.com     | 9000000012 |   27 | Indian      | 2026-04-15        |
|         13 | Daniel Garcia | daniel@gmail.com   | 9000000013 |   34 | Spanish     | 2026-04-20        |
|         14 | Meera Iyer   | meera.visitor@gmail.com | 9000000014 |   29 | Indian      | 2026-04-25        |
+------------+--------------+---------------------+------------+------+-------------+-------------------+
14 rows in set (0.00 sec)
```

SELECT , WHERE , ORDER BY, GROUP BY , aggregate functions, joins, subqueries:

```
mysql> SELECT artifact_id, artifact_name, estimated_value
    -> FROM ARTIFACT
    -> WHERE estimated_value > 300000;
+-------------+----------------------+-----------------+
| artifact_id | artifact_name        | estimated_value |
+-------------+----------------------+-----------------+
|           2 | Granite Temple Statue |       320000.00 |
|           7 | Royal Throne          |       550000.00 |
|           9 | Mythological Painting |       450000.00 |
|          10 | Bharat Mata Painting  |       500000.00 |
|          11 | Village Life Painting |       350000.00 |
|          13 | Temple Bronze Idol    |       400000.00 |
|          17 | Ancient Bronze Vessel |       600000.00 |
|          19 | Ancient Gold Coin     |       750000.00 |
+-------------+----------------------+-----------------+
8 rows in set (0.00 sec)
```

```
mysql> SELECT creator_id, creator_name, birth_year
    -> FROM CREATOR
    -> ORDER BY birth_year ASC;
+------------+--------------------------+------------+
| creator_id | creator_name             | birth_year |
+------------+--------------------------+------------+
|          9 | Ancient Metalworker      |       1100 |
|          2 | Unknown Ancient Sculptor |       1200 |
|          8 | Unknown Temple Artisan   |       1400 |
|          4 | Unknown Manuscript Artist |       1600 |
|         10 | Royal Coin Maker         |       1650 |
|          3 | Mysuru Royal Artisan     |       1750 |
|          1 | Ravi Varma               |       1848 |
|          5 | Abanindranath Tagore     |       1871 |
|          6 | Nandalal Bose            |       1882 |
|          7 | Jamini Roy               |       1887 |
+------------+--------------------------+------------+
10 rows in set (0.00 sec)
```

```
mysql> SELECT artifact_type, COUNT(*) AS total_artifacts
    -> FROM ARTIFACT
    -> GROUP BY artifact_type;
+--------------------+-----------------+
| artifact_type      | total_artifacts |
+--------------------+-----------------+
| Sculpture          |               3 |
| Manuscript         |               1 |
| Document           |               1 |
| Furniture          |               3 |
| Weapon             |               1 |
| Painting           |               4 |
| Religious Artifact |               1 |
| Craft              |               1 |
| Textile            |               1 |
| Vessel             |               1 |
| Terracotta         |               1 |
| Coin               |               2 |
+--------------------+-----------------+
12 rows in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) AS total_artifacts
    -> FROM ARTIFACT;
+-----------------+
| total_artifacts |
+-----------------+
|              20 |
+-----------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(estimated_value) AS average_artifact_value
    -> FROM ARTIFACT;
+------------------------+
| average_artifact_value |
+------------------------+
|          312750.000000 |
+------------------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT
    ->     A.artifact_id,
    ->     A.artifact_name,
    ->     C.collection_name
    -> FROM ARTIFACT A
    -> INNER JOIN COLLECTION C
    -> ON A.collection_id = C.collection_id;
+-------------+--------------------------+--------------------------+
| artifact_id | artifact_name            | collection_name          |
+-------------+--------------------------+--------------------------+
|           1 | Ancient Stone Sculpture  | Ancient Indian Art       |
|           2 | Granite Temple Statue    | Ancient Indian Art       |
|           3 | Royal Manuscript         | Medieval Manuscripts     |
|           4 | Historical Document      | Medieval Manuscripts     |
|           5 | Royal Wooden Chest       | Royal Heritage           |
|           6 | Royal Sword              | Royal Heritage           |
|           7 | Royal Throne             | Mysuru Palace Collection |
|           8 | Royal Jewellery Box      | Mysuru Palace Collection |
|           9 | Mythological Painting    | Indian Paintings         |
|          10 | Bharat Mata Painting     | Indian Paintings         |
|          11 | Village Life Painting    | Indian Paintings         |
|          12 | Folk Art Composition     | Modern Art               |
|          13 | Temple Bronze Idol       | Temple Artifacts         |
|          14 | Temple Stone Panel       | Temple Artifacts         |
|          15 | Traditional Wooden Doll  | Traditional Crafts       |
|          16 | Handwoven Textile        | Traditional Crafts       |
|          17 | Ancient Bronze Vessel    | Archaeological Collection |
|          18 | Ancient Terracotta Figure | Archaeological Collection |
|          19 | Ancient Gold Coin        | Coins and Currency       |
|          20 | Silver Royal Coin        | Coins and Currency       |
+-------------+--------------------------+--------------------------+
20 rows in set (0.00 sec)
```

```
mysql> SELECT artifact_id, artifact_name, estimated_value
    -> FROM ARTIFACT
    -> WHERE estimated_value >
    ->       (SELECT AVG(estimated_value)
    ->        FROM ARTIFACT);
+-------------+----------------------+-----------------+
| artifact_id | artifact_name        | estimated_value |
+-------------+----------------------+-----------------+
|           2 | Granite Temple Statue |       320000.00 |
|           7 | Royal Throne          |       550000.00 |
|           9 | Mythological Painting |       450000.00 |
|          10 | Bharat Mata Painting  |       500000.00 |
|          11 | Village Life Painting |       350000.00 |
|          13 | Temple Bronze Idol    |       400000.00 |
|          17 | Ancient Bronze Vessel |       600000.00 |
|          19 | Ancient Gold Coin     |       750000.00 |
+-------------+----------------------+-----------------+
8 rows in set (0.00 sec)
```

```
mysql> SELECT artifact_id, artifact_name, estimated_value
    -> FROM ARTIFACT
    -> WHERE estimated_value =
    ->       (SELECT MAX(estimated_value)
    ->        FROM ARTIFACT);
+-------------+------------------+-----------------+
| artifact_id | artifact_name    | estimated_value |
+-------------+------------------+-----------------+
|          19 | Ancient Gold Coin |       750000.00 |
+-------------+------------------+-----------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT artifact_type, COUNT(*) AS total_artifacts
    -> FROM ARTIFACT
    -> GROUP BY artifact_type
    -> HAVING COUNT(*) > 1;
+---------------+-----------------+
| artifact_type | total_artifacts |
+---------------+-----------------+
| Sculpture     |               3 |
| Furniture     |               3 |
| Painting      |               4 |
| Coin          |               2 |
+---------------+-----------------+
4 rows in set (0.00 sec)
```

Triggers:

Trigger to automatically calculate material cost

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_material_before_insert
    -> BEFORE INSERT ON RESTORATION_MATERIAL
    -> FOR EACH ROW
    -> BEGIN
    ->     SET NEW.total_cost = NEW.unit_cost * NEW.quantity_used;
    -> END$$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> INSERT INTO RESTORATION_MATERIAL
    -> (material_id, material_name, material_type, supplier,
    ->  unit, unit_cost, quantity_used, total_cost)
    -> VALUES
    -> (11, 'Conservation Wax', 'Protective Material',
    ->  'Heritage Supplies', 'kg', 1500.00, 4.00, 0);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT material_id, material_name, unit_cost,
    ->        quantity_used, total_cost
    -> FROM RESTORATION_MATERIAL
    -> WHERE material_id = 11;
+-------------+------------------+-----------+---------------+------------+
| material_id | material_name    | unit_cost | quantity_used | total_cost |
+-------------+------------------+-----------+---------------+------------+
|          11 | Conservation Wax |   1500.00 |          4.00 |    6000.00 |
+-------------+------------------+-----------+---------------+------------+
1 row in set (0.00 sec)
```

Trigger to prevent an exhibition ending before it starts

```
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_exhibition_before_insert
    -> BEFORE INSERT ON EXHIBITION
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.end_date < NEW.start_date THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Exhibition end date cannot be before start date';
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> DELIMITER ;
mysql> INSERT INTO EXHIBITION
    -> (exhibition_id, museum_id, exhibition_name,
    ->  description, theme, start_date, end_date, status)
    -> VALUES
    -> (9, 1, 'Invalid Exhibition',
    ->  'Testing date validation',
    ->  'Test',
    ->  '2026-12-01',
    ->  '2026-10-01',
    ->  'Planned');
ERROR 1644 (45000): Exhibition end date cannot be before start date
```

Functions:

Function to calculate artifact age

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION fn_artifact_age(p_artifact_id INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE v_age INT;
    ->
    ->     SELECT TIMESTAMPDIFF(
    ->         YEAR,
    ->         creation_date,
    ->         CURDATE()
    ->     )
    ->     INTO v_age
    ->     FROM ARTIFACT
    ->     WHERE artifact_id = p_artifact_id;
    ->
    ->     RETURN v_age;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SELECT
    ->     artifact_id,
    ->     artifact_name,
    ->     fn_artifact_age(artifact_id) AS Artifact_Age
    -> FROM ARTIFACT;
+-------------+---------------------------+--------------+
| artifact_id | artifact_name             | Artifact_Age |
+-------------+---------------------------+--------------+
|           1 | Ancient Stone Sculpture   |          776 |
|           2 | Granite Temple Statue     |          726 |
|           3 | Royal Manuscript          |          376 |
|           4 | Historical Document       |          326 |
|           5 | Royal Wooden Chest        |          246 |
|           6 | Royal Sword               |          251 |
|           7 | Royal Throne              |          226 |
|           8 | Royal Jewellery Box       |          206 |
|           9 | Mythological Painting     |          136 |
|          10 | Bharat Mata Painting      |          121 |
|          11 | Village Life Painting     |          106 |
|          12 | Folk Art Composition      |           91 |
|          13 | Temple Bronze Idol        |          576 |
|          14 | Temple Stone Panel        |          526 |
|          15 | Traditional Wooden Doll   |          146 |
|          16 | Handwoven Textile         |          126 |
|          17 | Ancient Bronze Vessel     |          926 |
|          18 | Ancient Terracotta Figure |         1026 |
|          19 | Ancient Gold Coin         |          376 |
|          20 | Silver Royal Coin         |          326 |
+-------------+---------------------------+--------------+
20 rows in set (0.00 sec)
```

Function to calculate total restoration cost

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION fn_total_restoration_cost(p_artifact_id INT)
    -> RETURNS DECIMAL(12,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE v_total DECIMAL(12,2);
    ->
    ->     SELECT COALESCE(SUM(cost), 0.00)
    ->     INTO v_total
    ->     FROM RESTORATION_PROJECT
    ->     WHERE artifact_id = p_artifact_id;
    ->
    ->     RETURN v_total;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> SELECT
    ->     artifact_id,
    ->     artifact_name,
    ->     fn_total_restoration_cost(artifact_id) AS Total_Restoration_Cost
    -> FROM ARTIFACT;
+-------------+---------------------------+-------------------------+
| artifact_id | artifact_name             | Total_Restoration_Cost  |
+-------------+---------------------------+-------------------------+
|           1 | Ancient Stone Sculpture   |                    0.00 |
|           2 | Granite Temple Statue     |                    0.00 |
|           3 | Royal Manuscript          |                    0.00 |
|           4 | Historical Document       |                30000.00 |
|           5 | Royal Wooden Chest        |                55000.00 |
|           6 | Royal Sword               |                    0.00 |
|           7 | Royal Throne              |               120000.00 |
|           8 | Royal Jewellery Box       |                    0.00 |
|           9 | Mythological Painting     |                75000.00 |
|          10 | Bharat Mata Painting      |                    0.00 |
|          11 | Village Life Painting     |                    0.00 |
|          12 | Folk Art Composition      |                    0.00 |
|          13 | Temple Bronze Idol        |                65000.00 |
|          14 | Temple Stone Panel        |                70000.00 |
|          15 | Traditional Wooden Doll   |                    0.00 |
|          16 | Handwoven Textile         |                45000.00 |
|          17 | Ancient Bronze Vessel     |                    0.00 |
|          18 | Ancient Terracotta Figure |                38000.00 |
|          19 | Ancient Gold Coin         |                    0.00 |
|          20 | Silver Royal Coin         |                    0.00 |
+-------------+---------------------------+-------------------------+
20 rows in set (0.00 sec)
```

Procedures:

Procedure to display all artifacts of a particular status

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE sp_artifacts_by_status(IN p_status VARCHAR(50))
    -> BEGIN
    ->     SELECT
    ->         artifact_id,
    ->         artifact_name,
    ->         artifact_type,
    ->         estimated_value,
    ->         status
    ->     FROM ARTIFACT
    ->     WHERE status = p_status
    ->     ORDER BY artifact_name;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL sp_artifacts_by_status('Needs Restoration');
+-------------+-----------------------+---------------+-----------------+--------------------+
| artifact_id | artifact_name         | artifact_type | estimated_value | status             |
+-------------+-----------------------+---------------+-----------------+--------------------+
|           2 | Granite Temple Statue | Sculpture     |       320000.00 | Needs Restoration |
|          16 | Handwoven Textile     | Textile       |        95000.00 | Needs Restoration |
|           9 | Mythological Painting | Painting      |       450000.00 | Needs Restoration |
|           7 | Royal Throne          | Furniture     |       550000.00 | Needs Restoration |
+-------------+-----------------------+---------------+-----------------+--------------------+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.03 sec)
```

Procedure to display restoration projects for an artifact

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE sp_artifact_restoration_projects(IN p_artifact_id INT)
    -> BEGIN
    ->     SELECT
    ->         project_id,
    ->         project_name,
    ->         objective,
    ->         start_date,
    ->         end_date,
    ->         status,
    ->         cost
    ->     FROM RESTORATION_PROJECT
    ->     WHERE artifact_id = p_artifact_id
    ->     ORDER BY start_date;
    -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CALL sp_artifact_restoration_projects(4);
+------------+-------------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
| project_id | project_name                  | objective                                           | start_date | end_date   | status    | cost     |
+------------+-------------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
|          8 | Historical Document Preservation | Preserve and digitally archive the historical document. | 2026-01-10 | 2026-04-30 | Completed | 30000.00 |
+------------+-------------------------------+-----------------------------------------------------+------------+------------+-----------+----------+
1 row in set (0.00 sec)
```